MIL-L-37166
5 December 1974

MILITARY SPECIFICATION

# LIDOCAINE HYDROCHLORIDE SOLUTION, Viscous, 2%, 100 cc

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers Lidocaine Hydrochloride Solution, Viscous, 2%, 100 cc. (See 6.2).

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

| | |
|---|---|
| PPP-C-00186B | Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals. |

STANDARDS

Military

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6505

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN PHARMACEUTICAL ASSOCIATION PUBLICATION

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

U.S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE,
FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and
Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, D.C. 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

Pharmacopeia of the United States.

(Application for copies should be addressed to the Mack Publishing Company, Easton, Pa. 18042.)

3. REQUIREMENTS

3.1 Material. Shall be a thick, viscid liquid containing the following:

| | | | |
|---|---|---|---|
| Lidocaine Hydrochloride (anhydrous basis) | 20.00 | mg/ml |
| Methylparaben | 0.7 | mg/ml |
| Propylparaben | 0.3 | mg/ml |
| Sodium Saccharin | 0.75 | mg/ml |
| Purified Water q.s. ad | 1.0 | ml |

and a suitable flavoring agent, Sodium Carboxymethylcellulose as a thickening agent, and Sodium Hydroxide to adjust the pH.

3.1.1 Assay for lidocaine hydrochloride. The finished solution shall contain not less than 95.0 percent and not more than 105.0 percent of the required amount of lidocaine hydrochloride, when determined as specified in 4.4.1.



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3.1.2 Identity of lidocaine hydrochloride. The presence of lidocaine hydrochloride in the finished solution shall be determined as specified in 4.4.2.

3.2 Ingredients.

3.2.1 Lidocaine hydrochloride, methylparaben, propylparaben, sodium carboxymethylcellulose, purified water, and sodium hydroxide. The lidocaine hydrochloride, methylparaben, propylparaben, sodium carboxymethylcellulose, purified water and sodium hydroxide, used in the manufacture of the solution, shall be in accordance with the tests, standards, and requirements of the U.S.P., including any supplements or revisions thereto.

3.2.2 Sodium saccharin. The sodium saccharin used in the manufacture of the solution shall be in accordance with the tests, standards, and requirements of the N.F., including any supplements or revisions thereto.

3.2.3 All other ingredients entering into the solution shall be of U.S.P. or N.F. quality, or if not included in either of these compendia, shall be suitable for use in this solution.

3.3 Delivery. Not more than 4 months shall have elapsed from the date of manufacture of the product to the date of delivery to the Government.

3.4 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.



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4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For the purpose of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of finished solution produced from one formulation, subjected to the same compounding and manufacturing operations, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with the procedures set forth in Military Standard MIL-STD-105, with an acceptable quality Level (AQL) of 1.0 percent defective for major defects and 2.5 percent defective for minor defects.

4.4 Tests. Tests and examinations shall be performed to determine compliance with all requirements of this document and all referenced documents.

4.4.1 Assay for lidocaine hydrochloride. Accurately transfer 5 ml of the viscous solution into a separatory funnel containing 15 ml of purified water. Mix thoroughly, add 5 ml ammonium hydroxide T.S., and extract by shaking with five 30-ml portions of chloroform. Filter each extracted portion through a cotton filter saturated with chloroform. Wash the filter with chloroform and combine the extracts and washings. Evaporate with a current of warm air, nearly to dryness, and add 50 ml of 0.01N sulfuric acid. Heat to drive off the residual chloroform. Cool and add several drops of a mixture consisting of 3 parts 0.1 percent alcoholic bromcresol green T.S. and 1 part 0.2 percent alcoholic methyl red T.S. Titrate the excess acid with 0.01N sodium hydroxide to a red violet colored endpoint. Calculate the quantity of lidocaine hydrochloride per ml of solution.

4.4.1.1 As an alternate to the above procedure, the assay procedure in the U.S.P. monograph for Lidocaine Hydrochloride Jelly shall be used.

4.4.2 Identity of lidocaine hydrochloride. Pipet into a separator containing 10 to 15 ml of purified water, a volume of solution equivalent to about 300 mg of lidocaine hydrochloride. Mix thoroughly, add 4 ml of ammonia T.S. and extract with four 15-ml portions of chloroform. Combine the chloroform extracts, and evaporate with the aid of warm air to dryness. Redissolve the crystals in solvent hexane, evaporate with the aid of warm air, and dry the residue in vacuum over silica gel for 24 hours. The lidocaine so obtained melts between 66° and 69° C., and shall comply with Identification Tests A and B in the "Lidocaine" monograph of the U.S.P.



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4.5 Examination of Preparation for Delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table I. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The Inspection Level shall be S-2 and the Acceptable Quality Level shall be 4.0 defects per hundrd units.

Table I. Examination of Preparation for Delivery

| Examination | Defect |
|---|---|
| Containers | Not as specified. |
| Quantity | Quantity in interior packages and shipping containers not as specified. |
| Materials | Any component missing or damage. |
| Workmanship | Inadequate application of components such as incomplete closure of container flaps, tape improperly applied, or containers distorted. |
| Markings | Omitted, incorrect, illegibly, improper size, location, sequence, or method of application. |

## 5. PREPARATION FOR DELIVERY

5.1 Shall be in accordance with all applicable requirements of Interim Federal Specification PPP-C-00186B, dated 15 June 1972, and as specified herein.

5.1.1 Immediate containers. Shall comply with the following: Shall be a commercially available bottle, with child-resistant safety closure; considered acceptable by the Food and Drug Administration (FDA) for the product contained therein and shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

5.2 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act, and shall include the information required below:



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5.2.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the date of manufacture.

5.2.2 Unit packages. Each unit package label shall bear the following:

(a) labeling information in accordance with commercial practice.

(b) the National Stock Number.

NOTE: The National Stock Number shall not be preceded by any prefix.

Strip labeling may be used.

Label adhesion test shall not apply.

If a unit package is not supplied, the National Stock Number shall appear on the immediate container label.

5.3 Packaging and Packing.

5.3.1 Unit of issue. One bottle containing 100 cc, as specified, constitutes one unit of issue.

5.3.2 Unit package. At the option of the contractor, each unit may be packaged as specified in 5.2.5 of PPP-C-00186B.

5.3.4 Procedure code. Procedure code No. 3 of PPP-C-00186B applies.

5.4 Marking.

5.4.1 Intermediate package. In paragraph 5.5.3 of PPP-C-00186B, add: "Date of manufacture shall be shown in lieu of date packed. Marking shall include the legends: 'SUBJECT TO DAMAGE BY FREEZING' and 'STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)'."

5.4.2 Exterior container. In paragraph 5.5.4 of PPP-C-00186B, add: "Date of manufacture shall be shown i lieu of date packed. Marking shall include the legends: 'SUBJECT TO DAMAGE BY FREEZING' and 'STORE AT CONTROLLED ROOM TEMPERATURE (59° - 86° F.)'."



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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) the National Stock Number.
(c) Selection of applicable levels of packaging and packing (See 5.3).

6.2 This specification covers the following item(s):

| National Stock No. | Item Identification |
|---|---|
| 6505-00-063-6197 | LIDOCAINE HYDROCHLORIDE SOLUTION, Viscous, 2%, 100 cc |

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1342

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.



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INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-L-37166

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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**DD FORM 1 JAN 72 1426** REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED

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